Algorithmic Recourse: from Counterfactual Explanations to Interventions

Karimi, Schölkopf, Valera (2020)

Reading Group: Ruelle & Lanzeray

Context:

- Machine learning increasingly used for consequential decisions
- Need for **explanations** and **actionable recommendations**

Key Questions:

1. Why was my loan denied?
2. What can I do to get approved?

Introduction: The Challenge of Algorithmic Recourse

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Core Contribution

Paradigm shift:

- From: Nearest counterfactual explanations
- To: Recourse through minimal interventions

Problem Setting

Mathematical Framework:

- $\mathbf{x}^F \in \mathcal{X}$: Factual instance (current state)
- $h : \mathcal{X} \to \{0, 1\}$: Fixed predictor
- $\mathbf{x}^{CFE}$: Counterfactual explanation
- **Goal:** $h(\mathbf{x}^{CFE}) \neq h(\mathbf{x}^F)$

Example:

- X_1: Annual salary (\$75,000)
- X_2: Bank balance (\$25,000)
- Predictor:
  $h = \text{sgn}(X_1 + 5 \cdot X_2 - \$225,000)$
- Current: $h(\mathbf{x}^F) = -1$ (loan denied)

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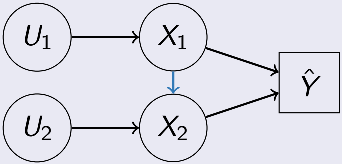
Graphical Model



Causal Model

$$X_1 := U_1$$
$$X_2 := f_2(X_1) + U_2$$
$$\hat{Y} = h(X_1, X_2)$$

Nearest Counterfactual Explanations

Optimization Problem (Wachter et al., 2017)

$$\mathbf{x}^{*CF} \in \arg\min_{\mathbf{x}} \mathrm{dist}(\mathbf{x}, \mathbf{x}^F) \quad \text{s.t.} \quad h(\mathbf{x}) \neq h(\mathbf{x}^F), \quad \mathbf{x} \in \mathcal{P}$$

$\mathcal{P}$: **Plausibility constraints**

- Salary must be within realistic bounds: $X_1 \in [\$20.000, \$500.000]$
- Bank balance non-negative and reasonable: $X_2 \in [\$0, \$1.000.000]$

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Critical Limitation

It assumes:

- Features are independent
- Direct manipulation of any feature is possible

CFE-based Recourse

Optimization Problem (Ustun et al., 2019)

$$\boldsymbol{\delta}^* \in \arg\min_{\boldsymbol{\delta}} \text{cost}(\boldsymbol{\delta}; \mathbf{x}^F) \quad \text{s.t.} \quad \begin{cases} h(\mathbf{x}^{CFE}) \neq h(\mathbf{x}^F) \\ \mathbf{x}^{CFE} = \mathbf{x}^F + \boldsymbol{\delta} \\ \mathbf{x}^{CFE} \in \mathcal{P}, \quad \boldsymbol{\delta} \in \mathcal{F} \end{cases}$$

- **Assumptions:**
  1. $\boldsymbol{\delta}^* = \mathbf{x}^{*CFE} - \mathbf{x}^F$ directly translates to actions
  2. $\text{dist}(\cdot,\cdot) \leftrightarrow \text{cost}(\cdot,\cdot)$ mapping exists

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- **Key Improvements:**
  1. Focus on actions $\boldsymbol{\delta}$
  2. $\mathcal{F}$: Feasibility constraints [1]

[1] example: age cannot decrease, gender immutable

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Problem Remains

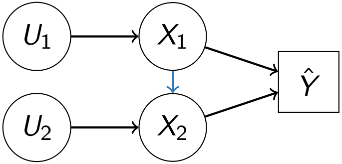
Still assumes feature independence! Changing one variable doesn't affect others.

[1] example: age cannot decrease, gender immutable

Causal Perspective: Actions as Interventions

Interventions:

- $do(X_i := a_i)$: Set X_i to value a_i
- Breaks incoming edges to X_i
- Propagates effects downstream



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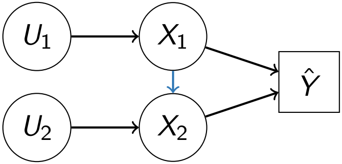

Theorem (Proposition 3.1 – When CFE Works)

CFE-based actions guarantee recourse **if and only if** *the intervened variables have no descendants.*

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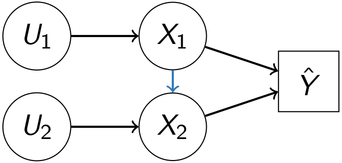

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Key Insight:

> *Intervening on X_1 affects downstream X_2, but CFE-based methods assume independence!*

Recourse through Minimal Interventions (MINT)

Optimization Problem (Karimi, Schölkopf, Valera, 2020)

$$\mathcal{A}^* \in \arg\min_{\mathcal{A}} \text{cost}(\mathcal{A}; \mathbf{x}^F) \quad \text{s.t.} \quad \begin{cases} h(\mathbf{x}^{SCF}) \neq h(\mathbf{x}^F) \\ \mathbf{x}^{SCF} = \mathbb{F}_{\mathcal{A}}(\mathbb{F}^{-1}(\mathbf{x}^F)) \\ \mathbf{x}^{SCF} \in \mathcal{P}, \quad \mathcal{A} \in \mathcal{F} \end{cases}$$

Key Components:

- $\mathcal{A}$: Set of structural interventions
- $\mathbb{F} : \mathbf{u} \mapsto \mathbf{x}^F$
- $\mathbb{F}_{\mathcal{A}} : \mathbf{u} \mapsto \mathbf{x}^{SCF}$ is the modified causal mechanism (set X_i to a_i)
- Directly specifies feasible actions $\mathcal{A} \in \mathcal{F}$

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Advantages:

- ✓ Respects causal structure
- ✓ Guarantees feasibility
- ✓ Provably optimal cost
- ✓ Models real-world constraints

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Theorem (Proposition 4.1 – Optimality)

For any CFE-based action $\mathcal{A}^{CFE}$ that achieves recourse:

$$cost(\mathcal{A}^*; \mathbf{x}^F) \leq cost(\mathcal{A}^{CFE}; \mathbf{x}^F)$$

How MINT Works: Abduction-Action-Prediction

1. **Abduction:** Infer the exogenous variables consistent with the factual instance:

$$u = \mathbb{F}^{-1}(\mathbf{x}^F)$$

2. **Action:** Modify the structural causal model according to the intervention set:

$$\mathbb{F}_{\mathcal{A}}$$

3. **Prediction:** Feed the same exogenous variables u into the modified model to obtain the structural counterfactual:

$$\mathbf{x}^{SCF} = \mathbb{F}_{\mathcal{A}}(u)$$

MINT Algorithm

Algorithm 1 MINT Recourse

Input: $\mathbf{x}^F$, SCM $\mathcal{M}$, predictor h
Initialize: $\mathcal{A} = \emptyset$
for each intervention set I **do**
  Compute $\mathbf{u} = \mathbb{F}^{-1}(\mathbf{x}^F)$ *// Abduction*
  for $i \in I$ **do**
    Choose candidate intervention values a_i
  end for
  Apply interventions $\mathcal{A}_I = do(\{X_i := a_i\}_{i \in I})$ *// Action*
  Compute $\mathbf{x}^{SCF} = \mathbb{F}_{\mathcal{A}}(\mathbf{u})$ *// Prediction*
  if $h(\mathbf{x}^{SCF}) \neq h(\mathbf{x}^F)$ **then**
    Update best if $\text{cost}(\mathcal{A}_I) < \text{cost}_{best}$
  end if
end for
Return: Optimal $\mathcal{A}^*$, $\mathbf{x}^{SCF}$

CFE vs MINT Approaches

CFE-based Approach

$$\boxed{\mathbf{x}^F = [75000, 25000]}$$

$$\boxed{h(\mathbf{x}^F) = -1 \text{ (denied)}}$$

$$\downarrow \; \delta^* = [0, +5000]$$

$$\boxed{\mathbf{x}^{CFE} = [75000, 30000]}$$

$$\boxed{h(\mathbf{x}^{CFE}) = 1 \text{ (approved)}}$$

Actions Required:

- Must increase X_2 by $5,000
- Assumes no dependency on X_1
- Ignores natural saving behavior

CFE vs MINT Approaches

MINT Approach

$$\boxed{\mathbf{x}^F = [75000, 25000]}$$

$$\boxed{h(\mathbf{x}^F) = -1 \text{ (denied)}}$$

$$\Big\downarrow \quad \mathcal{A}^* = do(X_1 := 85000)$$

$$\boxed{\mathbf{x}^{SCF} = [85000, 28000]}$$

$$\boxed{h(\mathbf{x}^{SCF}) = 1 \text{ (approved)}}$$

Actions Required:

- Increase X_1 by \$10,000
- X_2 automatically increases by \$3,000
- Leverages causal structure

CFE Actions Required:

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MINT Actions Required:

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Key Insight

$$\mathrm{cost}(\boldsymbol{\delta}^*; \mathbf{x}^F) \approx 2\,\mathrm{cost}(\mathcal{A}^*; \mathbf{x}^F)$$

MINT exploits the causal structure, reducing cost while producing feasible real-world interventions.

Limitations, Future Work & Critical Analysis

Current Limitations:

- × Requires true causal model
- × Static interventions only

Future Directions:

- → Learning causal structure from data
- → Partial causal knowledge

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Strengths:

- Solid theoretical foundation
- Clear improvement over existing methods
- Practical examples and demonstrations

Weaknesses:

- A bit technical to read with no prior knowledge
- Limited empirical evaluation
- Computational complexity not addressed